Nosocomial Infections and Nosocomial Pneumonia

Principles and Practice of Osteopathic Medicine II
Dr. Carl J. Abraham
Assistant Professor

Department of Clinical Medicine

Session Objectives

1. Define nosocomial infections and their prevalence.
2. Recognize the most common nosocomial etiological agents with emphasis on nosocomial pneumonia.
3. Explain the importance of infection control and major causes of nosocomial infections.
4. Describe what infection control programs do, including the importance of handwashing, isolation and precautions, surveillance/reporting and cluster/outbreak investigation.
5. Appreciate that choosing the appropriate isolation/precautions requires in-depth understanding of microbial transmission.
6. Discuss the challenges nosocomial diseases present for patient care.

Source: Course Syllabus

Scope of the problem

- Nosocomial: originating in and from the hospital (aka HAI – Hospital-acquired infection)
  - Healthcare-associated infections: acquired due to a connection with a healthcare network
- Major causes:
  - Failure of healthcare personnel to follow infection control guidelines
  - Multi-drug resistant (MDR) bacteria,
  - Increased # of immunocompromised patients
- HAIs complicate 4% of all hospitalizations in US
- Each year 2 million patients acquire a HAI
- In 1992, cost $4.5 billion

CDC

Multidrug-resistant pathogens NI

- Approximately 16% of NIs were associated with multidrug-resistant (MDR) pathogens
  - Methicillin-R *Staphylococcus aureus* (MRSA)*
  - Carbapenem-R *Enterobacteriaceae** (*A. baumannii, K. pneumoniae/oxytoca, E. coli)*
  - Multidrug R-*P. aeruginosa**
  - Multidrug R-*Acinetobacter**
  - Vancomycin-R *Enterococcus faecium* (VRE)
  - Extended-spectrum cephalosporin-R *K. pneumoniae*
  - Extended-spectrum cephalosporin-R *E. coli*
- MDR pathogen infections are associated with increased morbidity and mortality, need for broad spectrum antibiotic use and increased healthcare costs

***urgent threat**

https://www.cdc.gov/hai/surveillance/ar-patient-safety-atlas.html

Causes of Mortality, U.S.

Heart Disease	700,000
Stroke	160,000
Lung Cancer	157,000
HAI's	**100,000**
Colon Cancer	48,100
Motor Vehicle	45,000
Breast Cancer	40,600
Sepsis	32,000
AIDS	16,000

CDC

Most Common Nosocomial Infections, U.S. 2018

- Central line-associated bloodstream infections (CLA-BSI)
- Catheter-associated urinary tract infections (CA-UTI)
- Ventilator-associated pneumonia (VAP)
- Surgical Site Infections (SSI)
- Clostridium difficile infection (CDI)

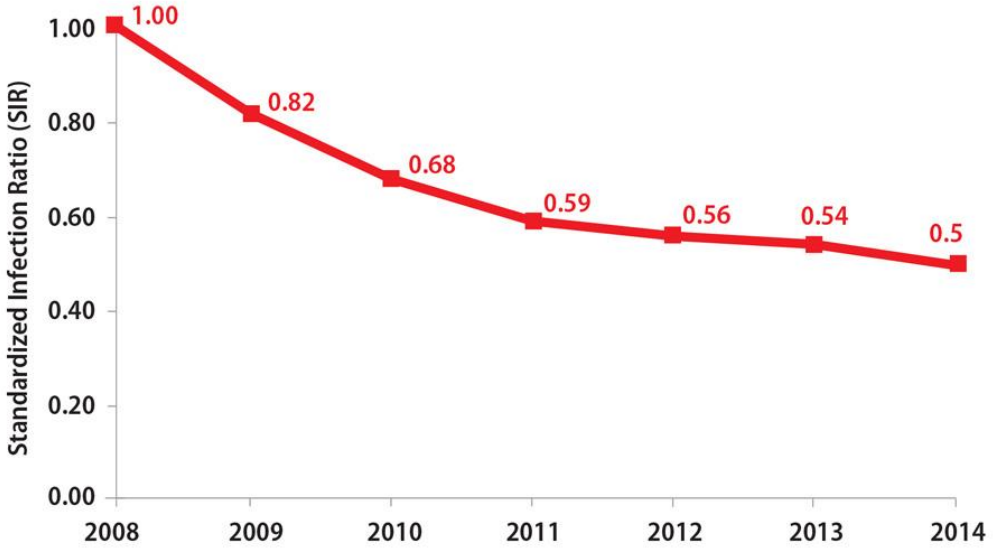
CDC

How to Prevent NIs: Infection Control!

- Reduce spread of infection from:
  - Patient to patient
  - Patient to worker
  - Worker to patient
  - Worker to worker
  - Visitor to patient to worker, etc.
- Identify (& report) nosocomial infections
  - Surveillance in (& interventions to reduce):
    - High risk areas (ICUs)
    - High risk infections (CLABSI, CAUTI, CDI, HCAP, VAP, SSI)
- Employee health issues (vaccination)
- ***Expanded Issues:*** *Reportable Diseases, Emerging pathogens/ bioterrorism response (H1N1), Antibiotic stewardship*

Nosocomial Infections, U.S. 2011

HAI Estimates Occurring in US Acute Care Hospitals, 2011	
Major Site of Infection	Estimated No.
Pneumonia	157,500
Gastrointestinal Illness	123,100
Urinary Tract Infections	93,300
Primary Bloodstream Infections	71,900
Surgical site infections from any inpatient surgery	157,500
Other types of infections	118,500
Estimated total number of infections in hospitals	**721,800**



CDC

NYIT College of Osteopathic Medicine

Birth of Infection Control

- Ignaz Semmelweis (1849)
  - In postpartum women with childbed fever (puerperal fever due to group A *Streptococci*) the mortality was higher when delivered by obstetricians & students and lower when delivered by midwives
  - Obstetricians & students frequently performed deliveries **after autopsies** (often on patients who died of childbed fever & other infectious diseases)
  - Instituted **hand washing** using a chlorine solution, **decreasing postpartum mortality from 11% to 1%**



Handwashing

- Single most effective intervention to reduce the cross transmission of nosocomial infections
- Must be "bacteriologically effective"
- Wash hands before any procedure in which gloves and forceps are necessary
- After contact with infected patient or one colonized with multidrug-resistant bacteria
- After touching infected material
- Use soap & water (preferably disinfectant soap)
- More prolonged & thorough scrub before surgery

Alcohol based hand sanitizers

- CDC hand antiseptic agents of choice
  - Recommended by CDC based on strong experimental, clinical, epidemiologic & microbiologic data
  - Antimicrobial superiority
    - Greater microbicidal effect
    - Prolonged residual effect
  - Ease of use & application



Isolation/Precautions

- Type 1: Prevent transmission of pathogen from patient to healthcare worker, e.g., influenza
  - Prevents
    - Healthcare worker acquisition
    - Transmission from patient to patient
- Type 2: Prevents transmission from healthcare worker to patient at risk, e.g., NICU, bone marrow transplant units
  - aka reverse precautions or protective isolation

Isolation/Precautions

- Standard precautions
  - Minimum set of infection prevention measures performed with all hospitalized patients
- Transmission-based precautions
  - Supplement standard precautions for patients known infected or colonized with certain transmissible pathogens

Standard Precautions

- Hand hygiene
- Personal protective equipment (e.g., gloves, gowns, facemasks) depending on procedure
- Respiratory hygiene & cough etiquette
- Safe injection practices
- Safe handling of potentially contaminated equipment or surfaces in the patient environment

CDC

Transmission based Precautions

- Contact precautions - direct/indirect contact with patient/environment
  - Placement: Single room/cohorting
  - PPE: Gloves, Gowns, dedicated equipment
  - Pathogens: MDROs such as MRSA & VRE
  - Enteric precautions – dedicated commode/bathroom
  - Pathogens: C. difficile

CDC

Transmission precautions

- Droplet precautions
  - Respiratory isolation to reduce transmission of large droplet nuclei
  - Patient places in isolation > 3 ft apart
  - No special air handling
  - PPE - regular surgical mask
  - Pathogens - B. pertussis, N. meningitidis, influenza virus

CDC, 2007 Isolation Precautions

Transmission precautions

- Airborne
  - Respiratory isolation to reduce transmission of small droplet nuclei
    - remain suspended in the air for long distances
  - Placement - Isolation, door closed (HEPA filtration/negative pressure)
  - PPE - N95 or higher respiratory mask
  - Pathogens - TB, measles, chicken pox, small pox

CDC, 2007 Isolation Precautions

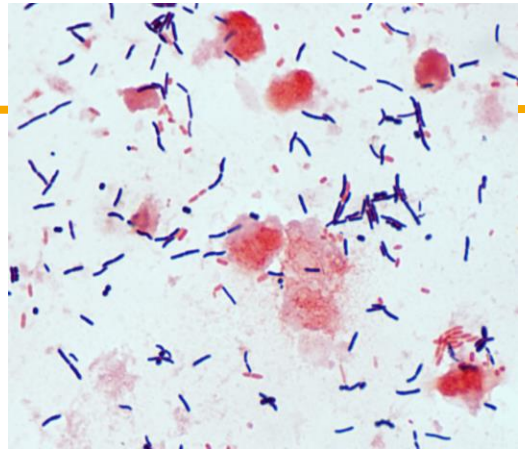
Most common NI pathogens

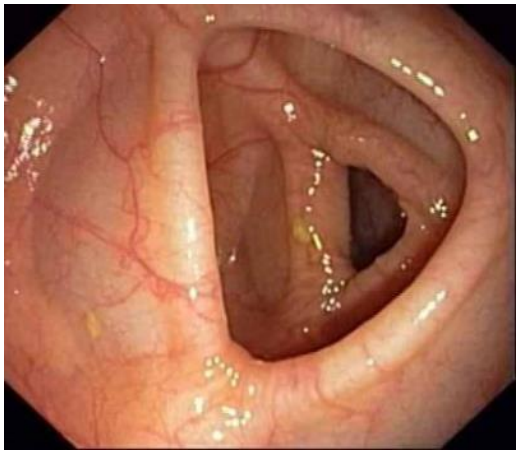
- *Clostridium difficile*
- Coagulase-negative *Staphylococci*
- *Staphylococcus aureus*
- *Enterococcus* spp.
- *Acinetobacter baumannii*
- *Escherichia coli*
- *Pseudomonas aeruginosa*
- *Klebsiella pneumoniae/oxytoca*
- *Candida* spp.
- *Enterobacter* spp.
- *Proteus* spp.
- *Serratia* spp.

https://www.cdc.gov/hai/surveillance/data-reports/2015-SIR-report.html

Clostridium difficile

- Gram +ve spore-forming rod
- Obligate anaerobe
- 2 toxins:
  - Toxin A: an enterotoxin that attracts neutrophils & causes cytokine production
  - Toxin B: a cytotoxin that causes necrosis
- The effect of both of these toxins is fluid leakage from the gut, diarrhea & formation of pseudomembranes



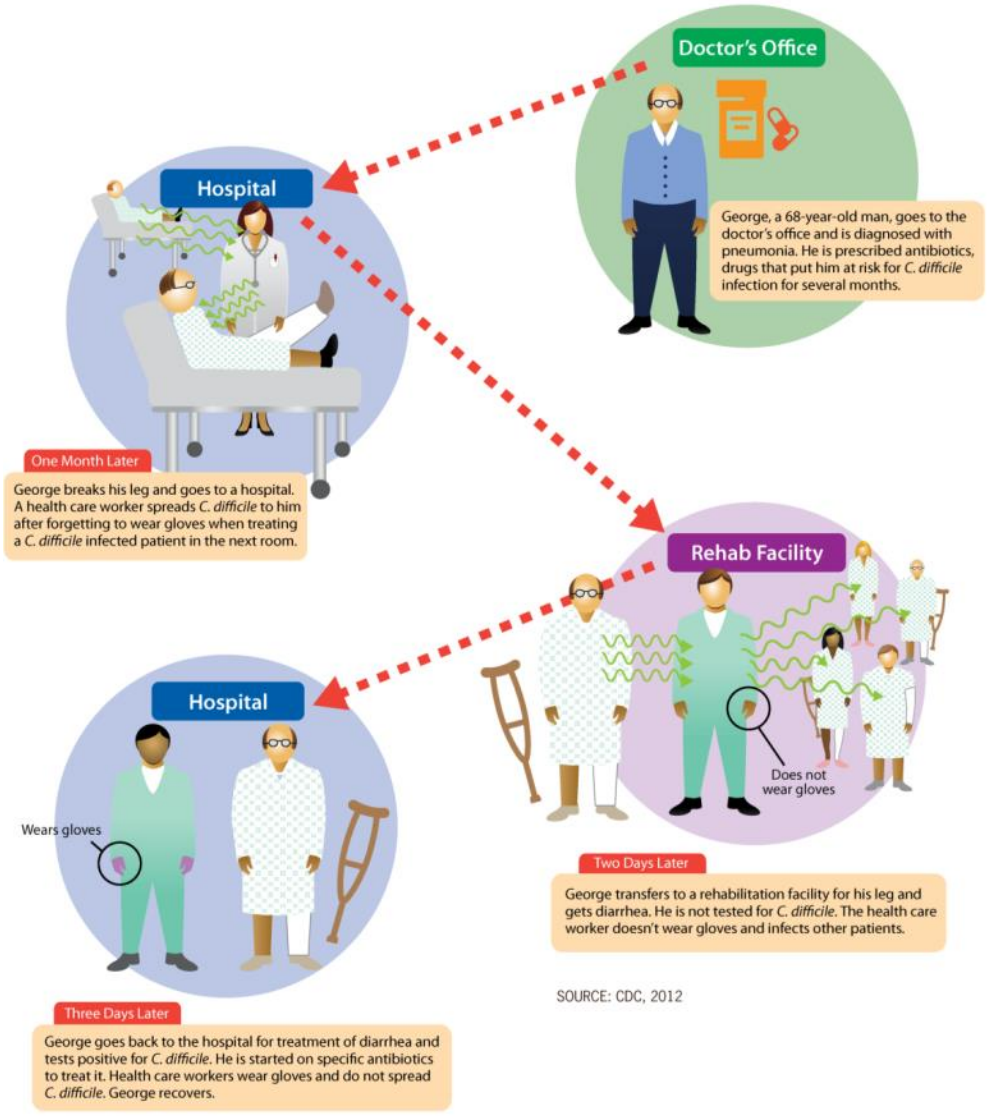




The rise of *C. difficile*

- Hypervirulent epidemic strain (B1/NAP1/027)
  - Implicated in outbreaks throughout the US, Canada & Europe
  - Greater toxin production
  - Greater antimicrobial resistance
  - Community spread
  - Control: infection control, environmental cleaning & antibiotic stewardship

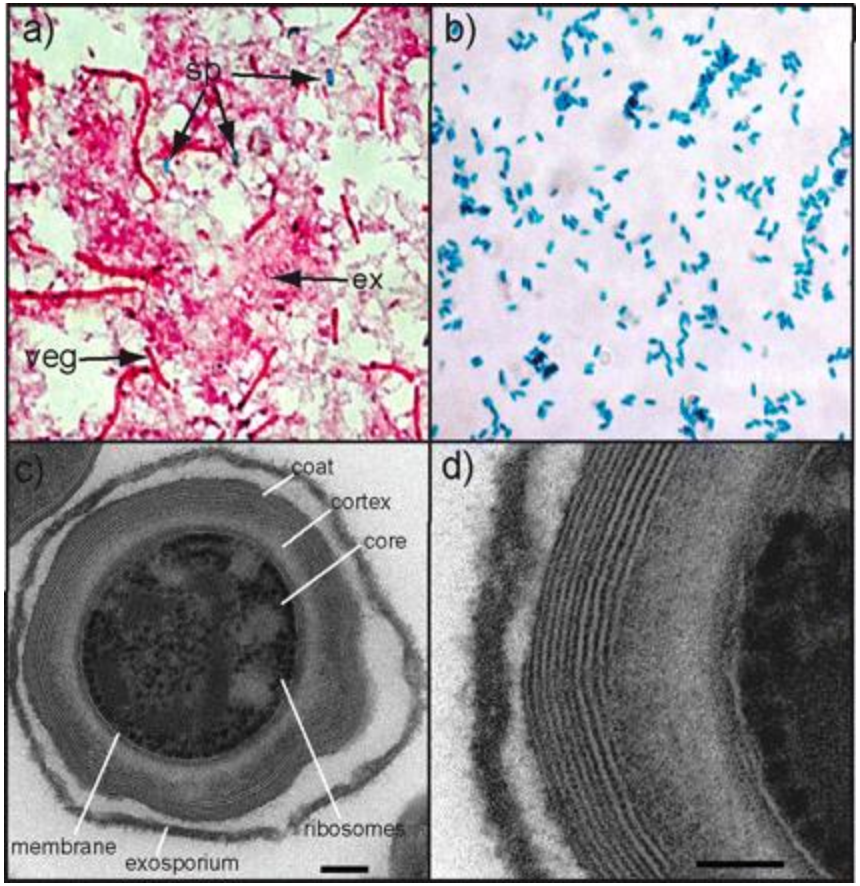
How *C. difficile* spreads?

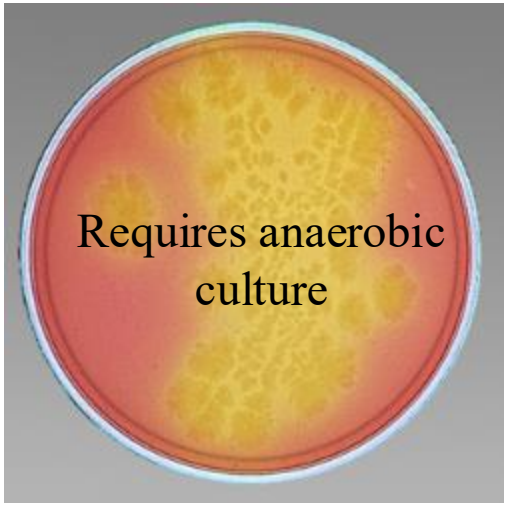


NYIT College of Osteopathic Medicine

Diagnosis of *C. difficile*

Gram-positive, spore-forming rods



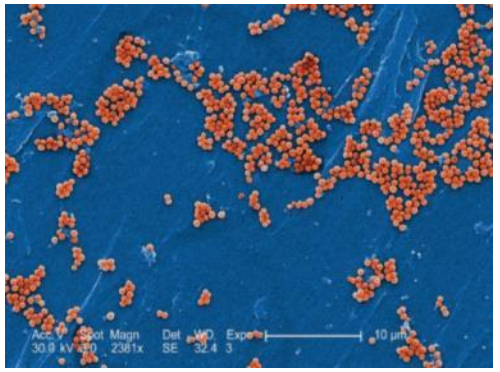




Stools positive for Toxin A (enterotoxin) and/or B (cytotoxin)

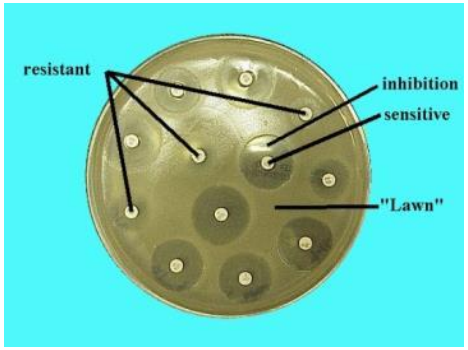
Staphylococci

- Gram +ve cocci
- Normal flora of humans:
  - Nasal membranes/skin
- Opportunistic infections:
  - Focal damage
    - 53% purulent wounds
    - 61% abscesses
  - Systemic disease
- Nosocomial infections
- MRSA is a strain resistant to the beta lactam antibiotic methicillin



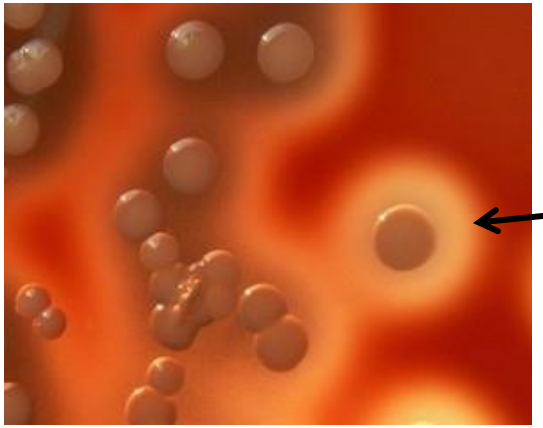






Characteristic of the genus *Staphylococcus*

- All members are highly resistant to drying, extremes in pH, & high temperatures (Biofilm formation)
- Increase in community acquired cases, with infections associated with schools, prisons, & shared athletic equipment
- MRSA was responsible for 94,000 life-threatening infections & nearly 19,000 deaths in the US
- *S. epidermidis* is a common cause of nosocomial infections in immunocompromised patients



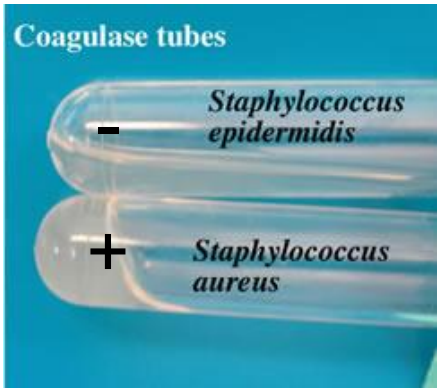
S. aureus is β-hemolytic on blood agar

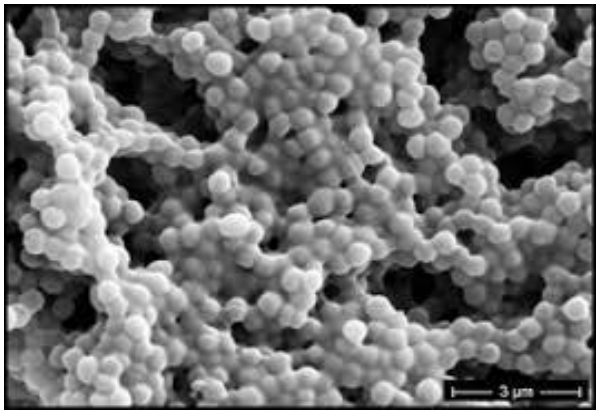
TABLE 32.1 Differentiation of staphylococcal species

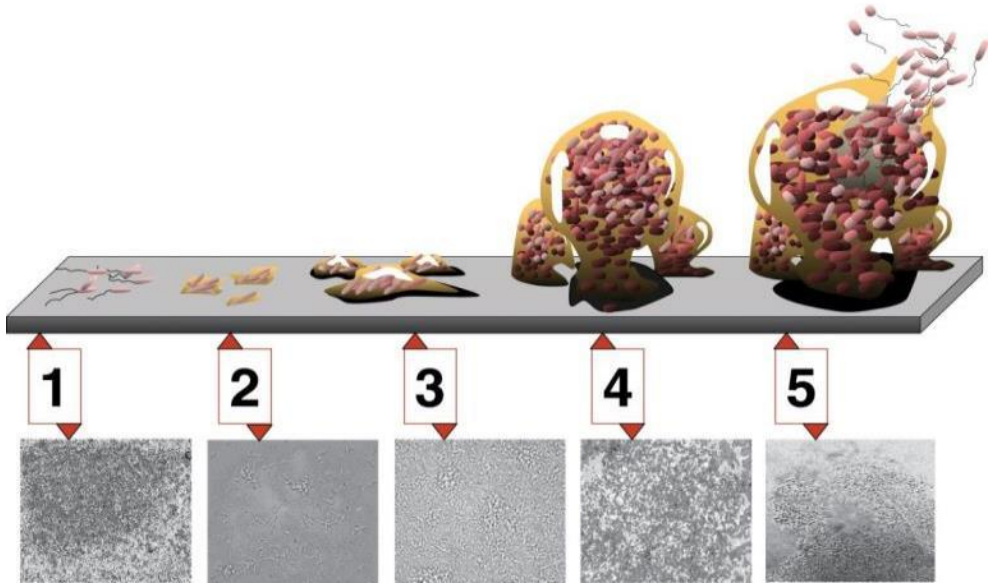
	S. aureus	*S. epidermidis*
Growth on media containing 7.5% NaCl	+	+
Beta hemolysis	+	–
Mannitol fermentation	+	–
Coagulation of plasma	+	–
Reaction to novobiocin	Sensitive	Sensitive

Staphylococcus aureus

- Catalase (+)
- Coagulase (+)
- Grows in high NaCl & ferment mannitol
- Biofilms (e.g. prosthetic devices)





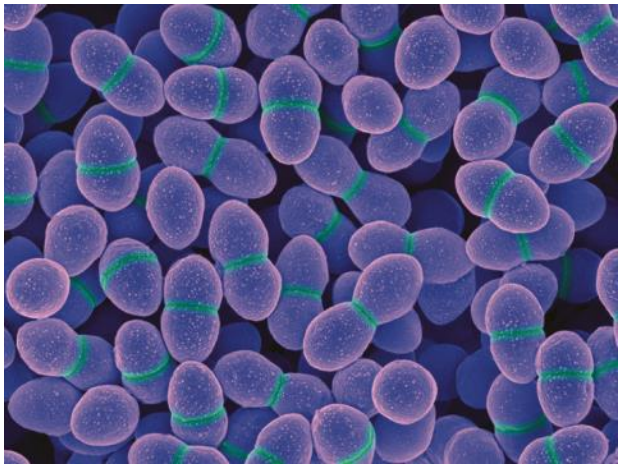


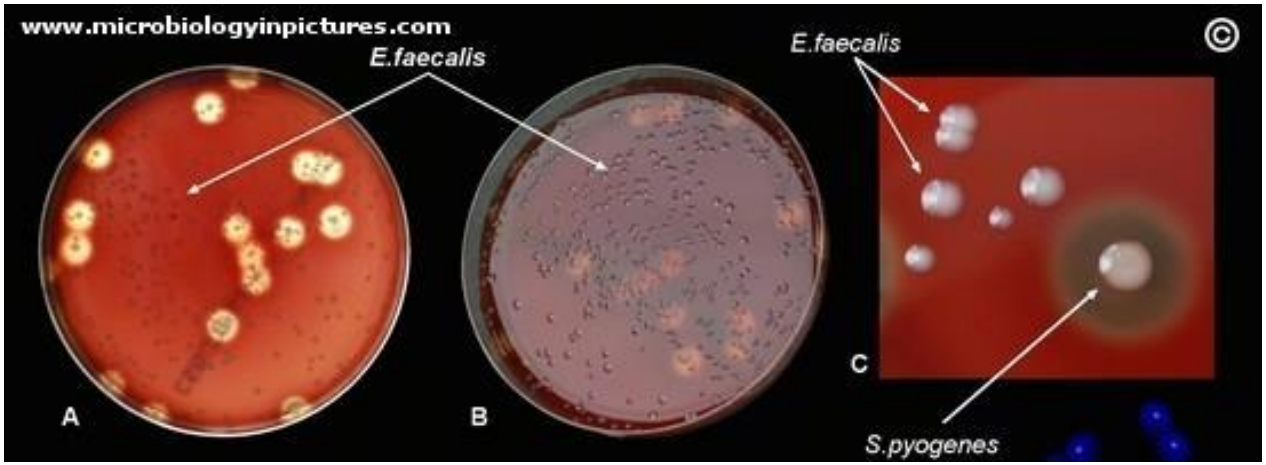
Enterococcus spp. (Group D *Streptococci*)

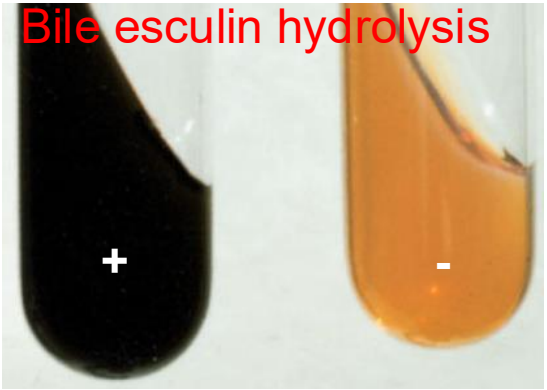
- Lancefield Group D (+) cell wall Ags (teichoic acid)
- Normal colonists of GI
- Transmitted person-to-person (direct contact)
- Occasionally on medical devises (fomites)
- Cause opportunistic urinary, wound, & skin infections, particularly in debilitated persons
- Cephalosporin and other antibiotics select for the strains
- Vancomycin-resistant *Enterococci* (VRE) are an important threat in nosocomial infections
  - 30% of *Enterococci* in ICUs are VRE

Enterococcus spp.

- Gram +ve cocci, chains
- Catalase -ve
- Grow in the presence of bile salts & 6.5% NaCl
- Grow well btw 10-45°C
- Hydrolyze esculin





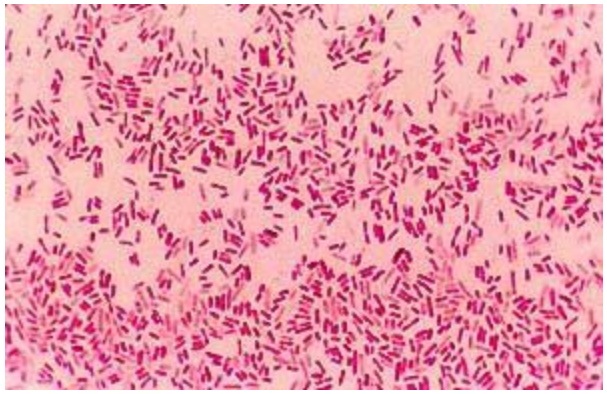


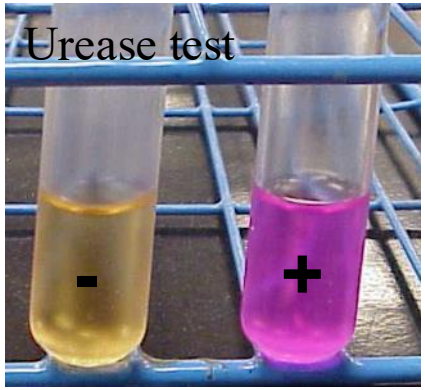
Enterococcus spp.

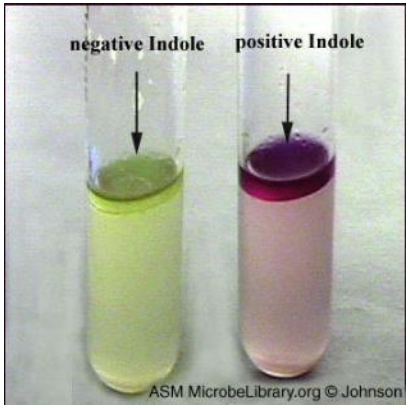
- *E. faecalis* is most common pathogen (85-90% of cases) but in HAI *E. faecium* is more common
- *E. faecium* more antimicrobial resistant than *E. faecalis* especially to vancomycin (multiple phenotypes)
- Intrinsic resistance to cephalosporins, penicillins, monobactams, aminoglycosides & fluoroquinolones
- Inhibited by ampicillin but not killed

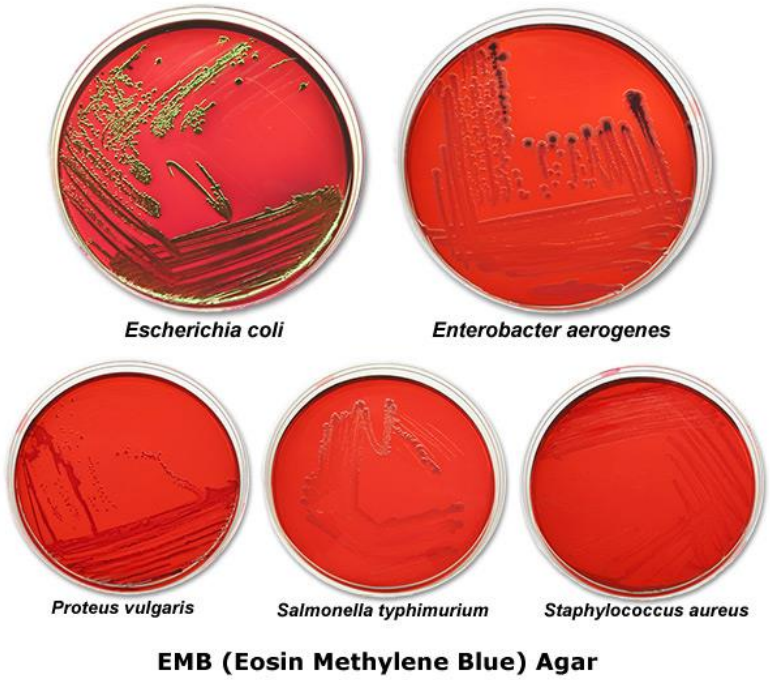
Escherichia coli

- Gram -ve rod
- UTIs, bacteremia, pneumonia, meningitis, & wound infections
- Ferments lactose (on EMB, black colonies and metallic green sheen or MAC agar)
- Indole +ve and urease -ve
- *E. coli* is the most abundant facultative anaerobe in the colon & feces
- 3 antigens (Ag) are used to type *E. coli*: O-Ag (LPS), H-Ag (flagella), and K (capsular). e.g. ***E. coli* O157H7**



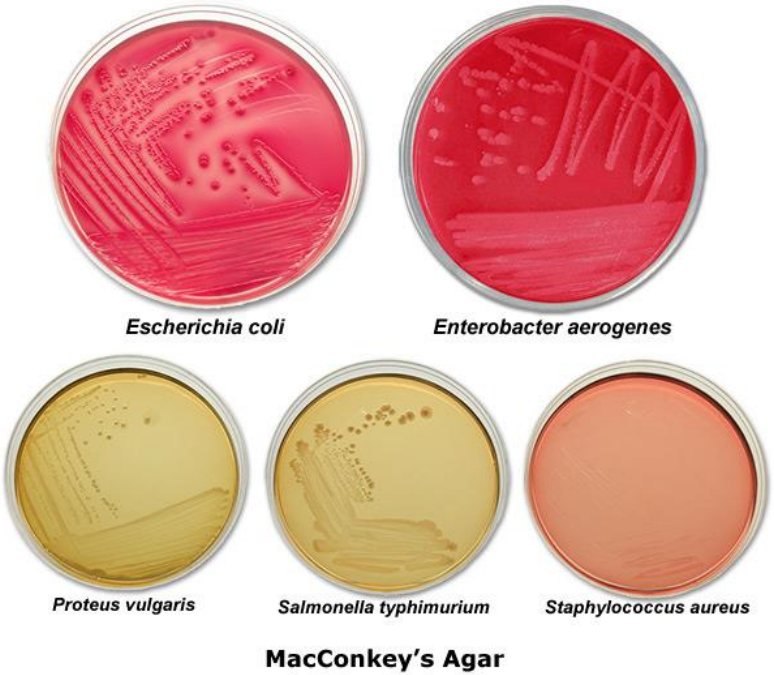






E. coli

- **MacConkey** agar (**MAC**) that contains crystal violet and bile salts which are toxic to Gram +ve bacteria, allowing only the growth of Gram -ves (**selective**). It is **differential** for lactose fermentation (**red/pink colonies**)



E. coli are indole +ve/urease -ve (rapid test) which differentiates them from other lactose-fermenting Gram -ve rods

- *Klebsiella*: indole -ve and urease +ve (rapid-test)
- *Enterobacter:* indole -ve and urease -ve (rapid-test)

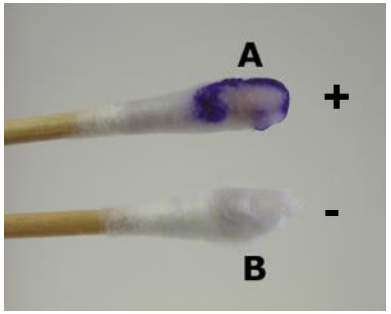
Pseudomonas species: characteristics

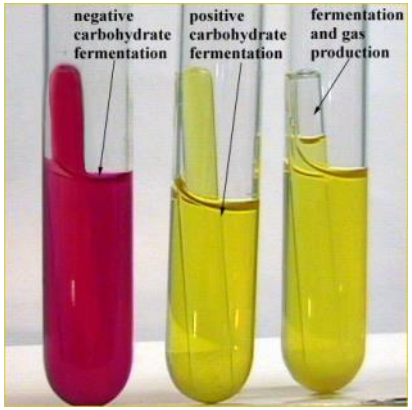
- *Enterobacteriaceae*
- Aerobic non-fermenting, actively motile Gram -ve rod
- Oxidase +ve
- Can survive in marginal environments: water, soil, vegetables/fruit, water taps & drains
- Pyocyanin pigment - blue green (fruity/grape odor)
- Resistant to commonly used antibiotics

Pyocyanin in culture



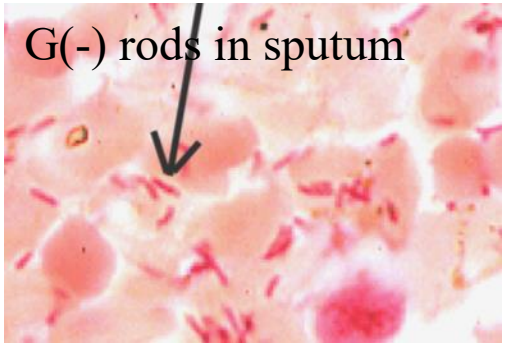
Oxidase-positive

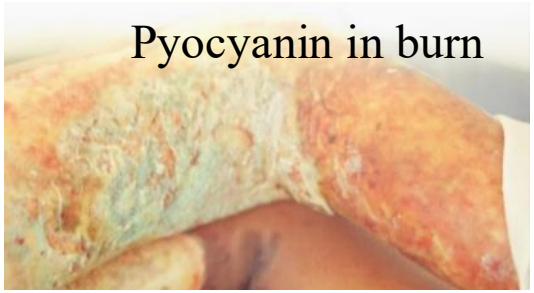




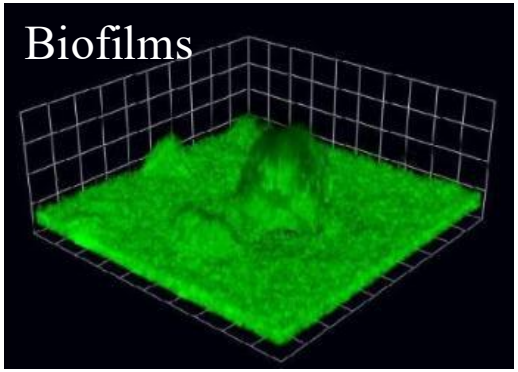
Pseudomonas species

- Can involve almost any part of body
- Especially a pathogen in immunosuppressed hosts- malignancy, diabetes, chemotherapy, other immunosuppression
  - Most common pathogen isolated from patients who have been hospitalized longer than 1 week
  - A frequent cause of nosocomial infections such as pneumonia, urinary tract infections (UTIs), & bacteremia
  - Chronic infection of the lower respiratory tract prevalent among patients with CF
  - Burn victims





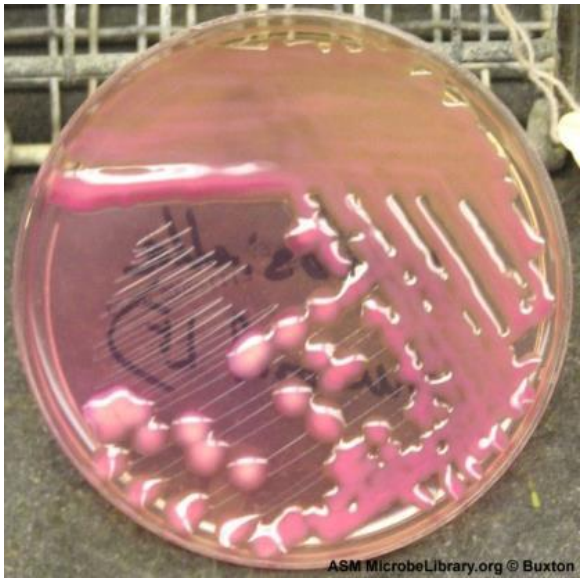


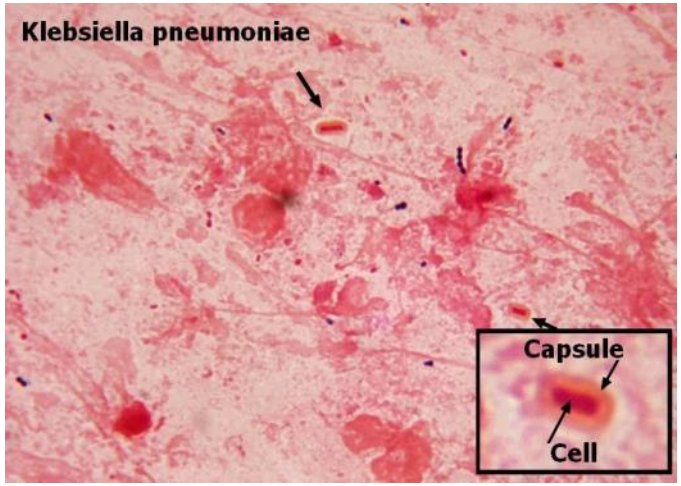


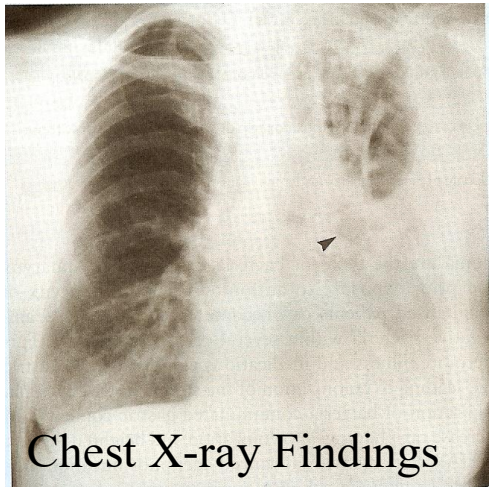
Klebsiella species: characteristics

- Genus *Klebsiella* (*Enterobacteriaceae*)
- Gram -ve
- Prominent polysaccharide capsule (accounts for large appearance on Gram stain & mucoid colonies)
- Lactose-fermentor, urease +ve, & indole -ve
- 2 types of Ags on cell surface:
    - LPS (O antigen) - 9 O antigens
    - Capsular polysaccharide (K antigen) - 77 K antigens
- 3 main species (7 similar DNA homology)
    - *K. pneumoniae* - most medically important
    - *K. oxytoca & K. rhinsclermatis*

Klebsiella species: characteristics






Chest X-ray Findings

Clinical syndromes-Pneumonia

- Ubiquitous in nature
- Humans can be colonized in skin, pharynx, GI tract (can colonize wounds & urine)
- Infection of the lungs (*K. pneumoniae* & *K. oxytoca*)
- Lobar pneumonia (usually upper lobe)
- Can cause destructive changes (abscess formation, cavitation, empyema & pleural adhesions)
- Necrosis, inflammation, hemorrhage & thick bloody mucous sputum

Klebsiella: HAIs

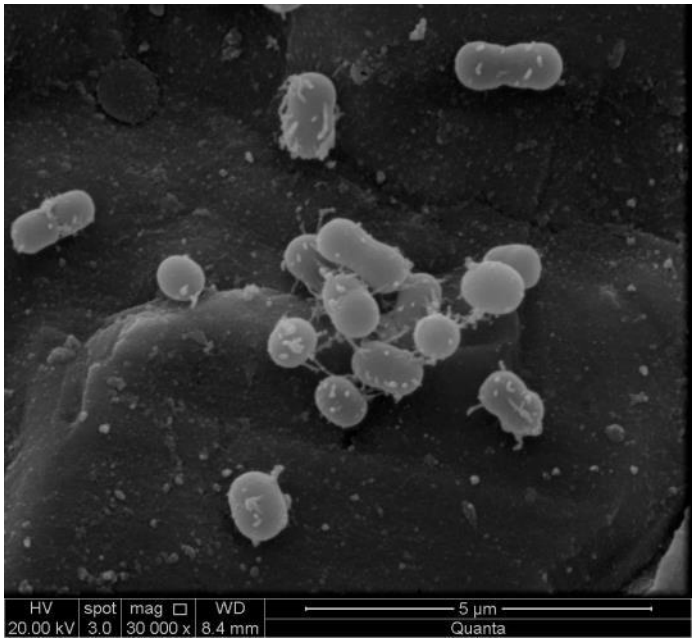
- Syndromes*:

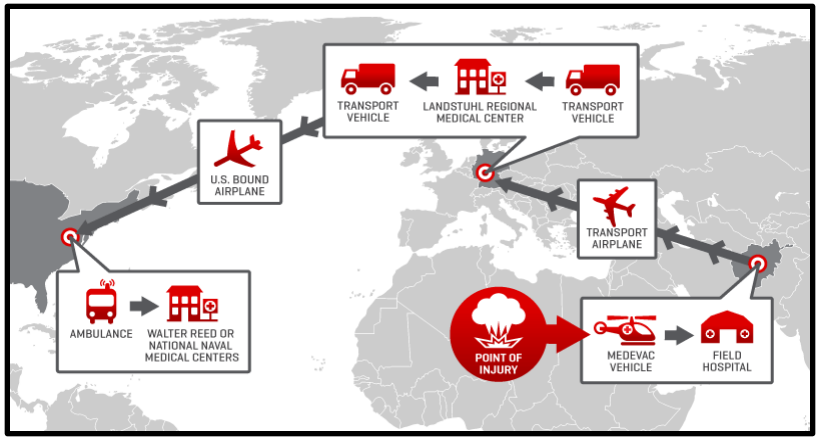
Pneumonias	Bacteremia
Thrombophlebitis	Urinary tract infection (UTI)
Cholecystitis	Diarrhea
Upper respiratory infection	Wound infection
Osteomyelitis	Meningitis

  - *Sepsis and shock may follow with entry of organism into blood from a focal source
- Increasing likelihood HAI with *Klebsiella* spp.:
  - Invasive devices
  - Contaminated respiratory support equipment
  - Use of urinary catheters
  - Use of antibiotics
- *K. oxytoca* implicated in neonatal bacteremia-premature infants and in neonatal ICUs

Acinetobacter baumannii

- Gram -ve coccobacillus
- Generally low virulence
- Opportunistic
- Adaptable growth requirements
- Aerobic, non-motile
- Biofilm formation
- Oxidase –ve & non-fermentor
- Nasopharyngeal transient flora
- Multi-drug resistance
- Most important complication in military wound victims

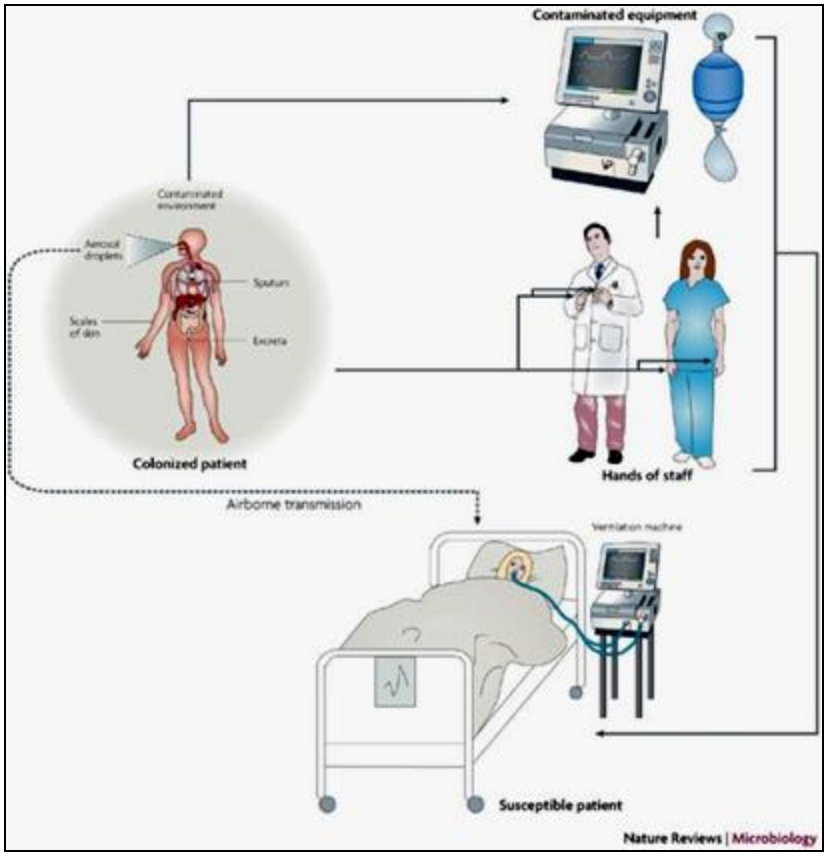




NYIT College of Osteopathic Medicine

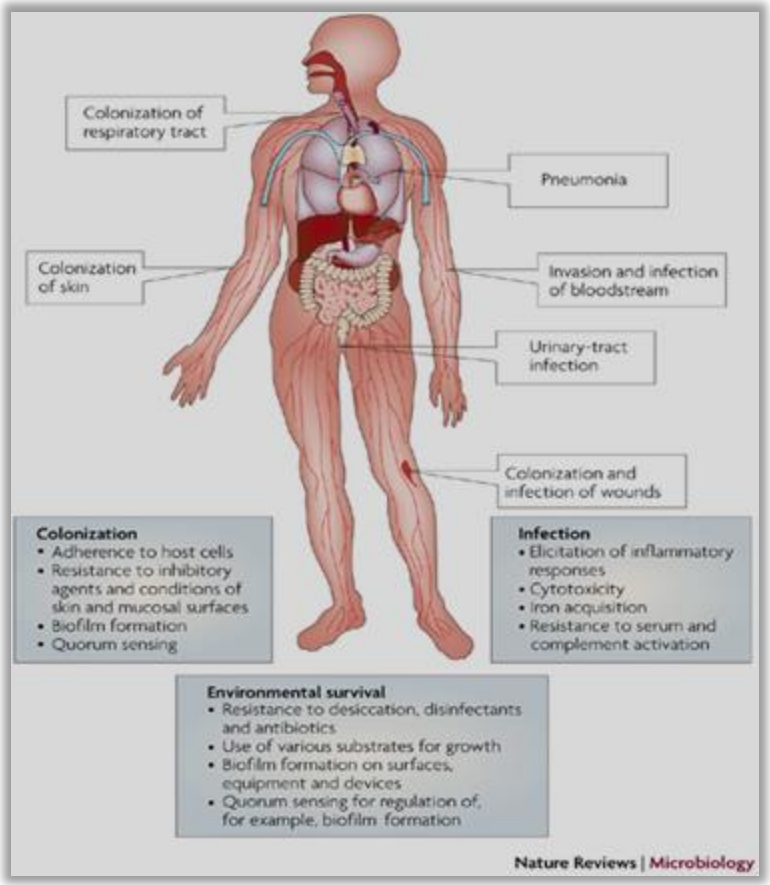
Transmission & prevalence

- Mostly found in nosocomial settings like I.V. solutions, urinary catheters, surgical drains or monitoring devices
- The bacteria can live up to 5 months on undisturbed surfaces
- Mostly enters in the body through open wounds, catheters & breathing tubes (respiratory tract)
- It affects immunocompromised individuals
- Community-acquired pneumonia affects alcoholics in tropical regions & frequently leads to systemic infection and has a mortality rate of ~50%



Colonization & Infection

- Infections usually occur in organs with high fluid content like respiratory tract, CSF, peritoneal fluid, urinary tract manifesting nosocomial pneumonia
- It can cause nosocomial meningitis, sepsis & UTI
- Also implicated in necrotizing fasciitis



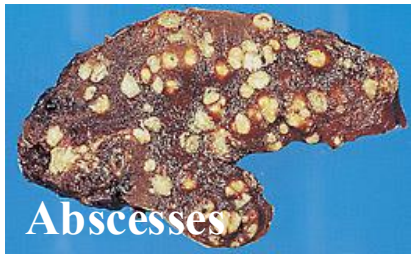


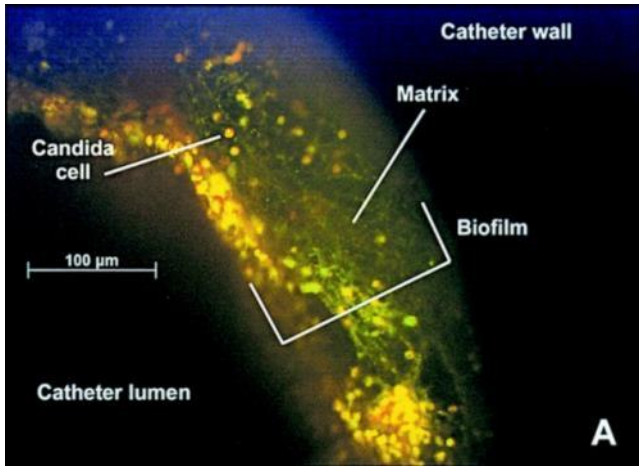


Candida species









- Most "important" fungal pathogen
- Associated with impaired immunity
- *C. albicans*: most common species (dimorphic)
- "non-*albicans*" *Candida*: increasing in incidence
- Spectrum of clinical syndromes: mild to fatal, dependent on host immune status
  - Disseminated, cutaneous and mucosal (e.g. oral thrush, vaginitis, etc.), UTIs, and neonatal candidiasis
- **Risk factors:** catheters, surgery, burns, prolonged antibiotic usage, cytotoxic therapy, parenteral hyper alimentation, pregnancy, diabetes

Candida: non-*albicans* species

Associated with bloodstream infections in North America

- Most common
  - *C. albicans*: germ tube formation
  - *C. glabrata*: not dimorphic; often resistant to azoles, echinocandin resistance appearing
  - *C. parapsilosis*: parenteral hyperalimentation; less susceptible to echinocandins
  - *C. tropicalis*: adheres to plastic
- *Less common*
  - *C. krusei*: resistance to azoles
  - *C. lusitaniae*: resistance to amphotericin B
  - *C. guilliermondii*: resistance to amphotericin B
  - *C. dubliniensis*: germ tube formation
  - *C. aurus*: panresistant

CDC

Nosocomial Pneumonia (NP)

- Definition:
  - The presence of a new pneumonia that occurs 48 hours after admission to the hospital that was not present on admission to the hospital
- Economic consequences
- Prolongs hospital stay 8-9 days
- Costs $1 billion/year

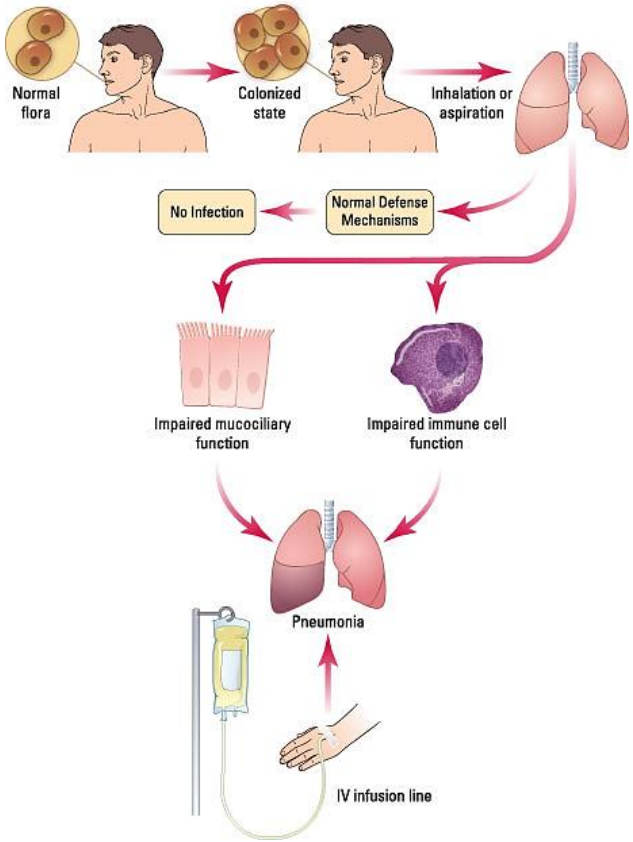
How is NP acquired?

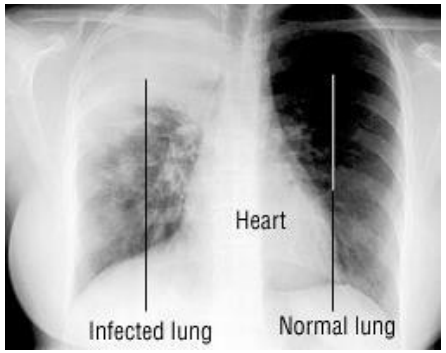
- **3 types:**

1. **Primary inhalation pneumonia** develops:
   - when microbes bypass normal respiratory defense mechanisms
   - when the patient inhales microbes that colonize the upper respiratory tract
   - when the patient is connected to respiratory support equipment (ICU)
2. **Aspiration pneumonia** occurs when the patient aspirates colonized upper respiratory tract secretions
3. **Hematogenously-acquired infections**
   - In bacterial nosocomial pneumonia, blood cultures frequently are positive if obtained early in the disease and if the patient is not already receiving antimicrobial therapy

NP

- The National Healthcare Safety Network (NHSN) reports that NP accounts for **31%** of all nosocomial infections in ICUs
- The mortality rate reaches 20–50%
- The increase in antibiotic-resistant pathogens in hospitals has contributed to the seriousness of these infections
- Respirator associated: *Klebsiella*, *Pseudomonas*, *Acinetobacter*, *Staphylococci*
- Non-respirator associated: *Legionella**





*= Can be part of the "Atypical Pneumonia" syndrome

Diagnosis, treatment & prevention

- Clinical diagnosis
  - fever, change in PO_2, change in sputum, CXR
- Microbiologic confirmation
  - Suctioned sputum sample
  - Bronchoscopy with brochoalveolar lavage
- Empiric antibiotic - clinical acumen
  - Tx based on previous cultures, usual hospital flora & susceptibilities
  - Sputum Gram stain
  - Colonization vs. infection
- Pulmonary hygiene
  - Change position every 2-hrs
    - Elevate head to 30-45 degrees
  - Deep breathing & incentive spirometry
  - Frequent suctioning
  - Bronchoscopy to remove mucous plugging

Summary

- Defined nosocomial infections and their prevalence.
- Reviewed the most common nosocomial etiological agents with emphasis on nosocomial pneumonia.
- Reviewed the importance of infection control and major causes of nosocomial infections.
- Reviewed infection control programs, including the importance of handwashing, isolation and precautions
- Reviewed appropriate isolation/precautions
- Reviewed the challenges nosocomial diseases present for patient care.

Lecture Feedback Form:

Please leave feedback (https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R)

Office Hours as stated in Canvas

cabrah04@nyit.edu